

MyShell: Interactive Linux Learning Environment

A Beginner-Centric Shell for Learning Linux and OS Concepts

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The Problem: Why Beginners Struggle with Linux

Too Many Commands

Beginners memorize syntax without grasping the underlying OS concepts that give it meaning.

Intimidating Environment

The blank terminal provides no guidance, no roadmap, and no safety net for first-time users.

Fragmented Learning

Learning, practice, and assessment are scattered across different tools with no unified experience.

No Progress Tracking

No clear learning roadmap, no immediate feedback, and no way to measure improvement over time.

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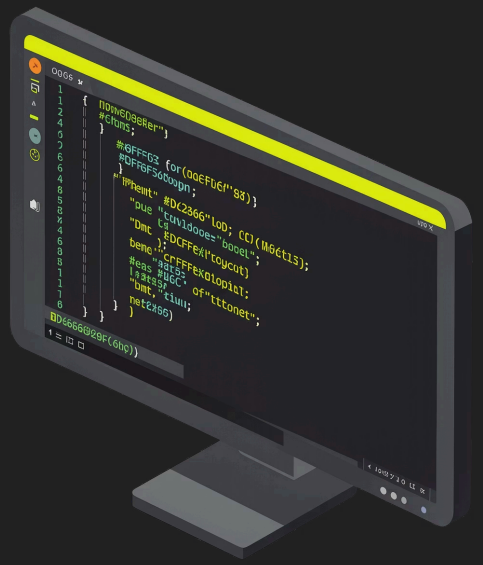
The Solution: MyShell Learning Workflow

INTEGRATED METHODOLOGY



Instead of separating learning, practice, and assessment, MyShell unifies them — commands are explained with context, students practice immediately within the shell, quizzes assess understanding in real-time, and progress is tracked at every stage.

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Technical Foundation

MyShell is built on a robust **C-based Linux shell implementation** using POSIX system calls — providing the technical backbone that the educational layer sits upon.

Language & OS

C · Linux/Ubuntu · GCC

Process Management

fork() · exec() · wait()

Shell Features

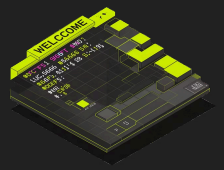
Pipes · I/O Redirection · Parsing

Persistence

JSON-based progress storage

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Current Implementation: Core Features



Learning Interface

- Beginner-friendly welcome screen
- Structured roadmap: ls → pwd → cd → ps → pipes
- Command lessons with OS concept explanations



Practice & Assessment

- Guided step-by-step practice activities
- Quiz system with instant scoring
- Progress tracking and persistence

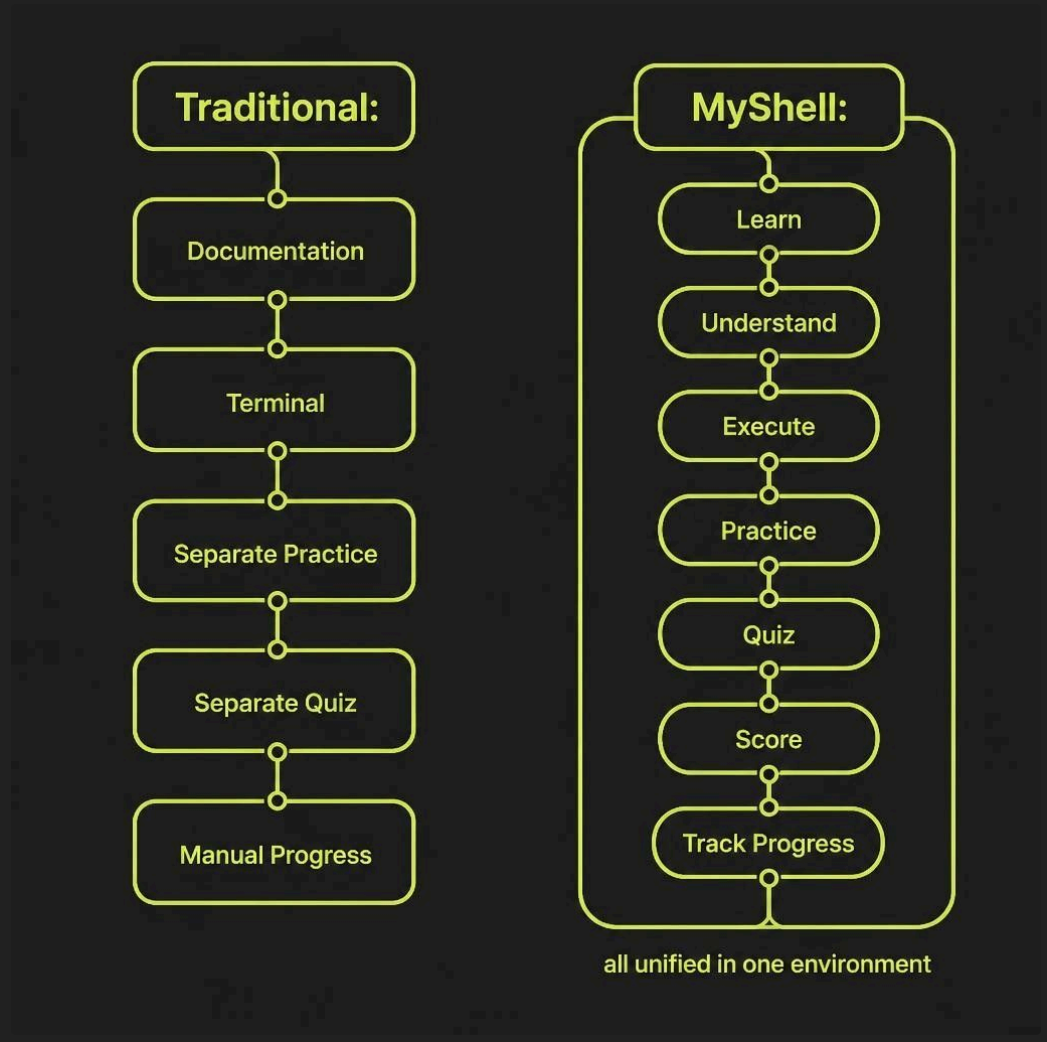


Shell Capabilities

- Linux command execution (ls, pwd, cd, ps)
- Pipe support — e.g., `ls | grep`
- Process management & background processes

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Novelty: Integrated Shell-Based Learning



Key Innovation

The novelty lies in the **integration**, not reinvention. MyShell unifies elements that have never been combined in a single shell environment:

- Linux shell implementation in C
- Beginner-oriented teaching methodology
- Command explanation with OS context
- Hands-on execution within the learning environment
- Practice activities, quizzes, and automated scoring
- Persistent progress tracking

The shell itself becomes the classroom — no switching between tools.

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Adaptability: Current and Future Scope

✓ Currently Implemented

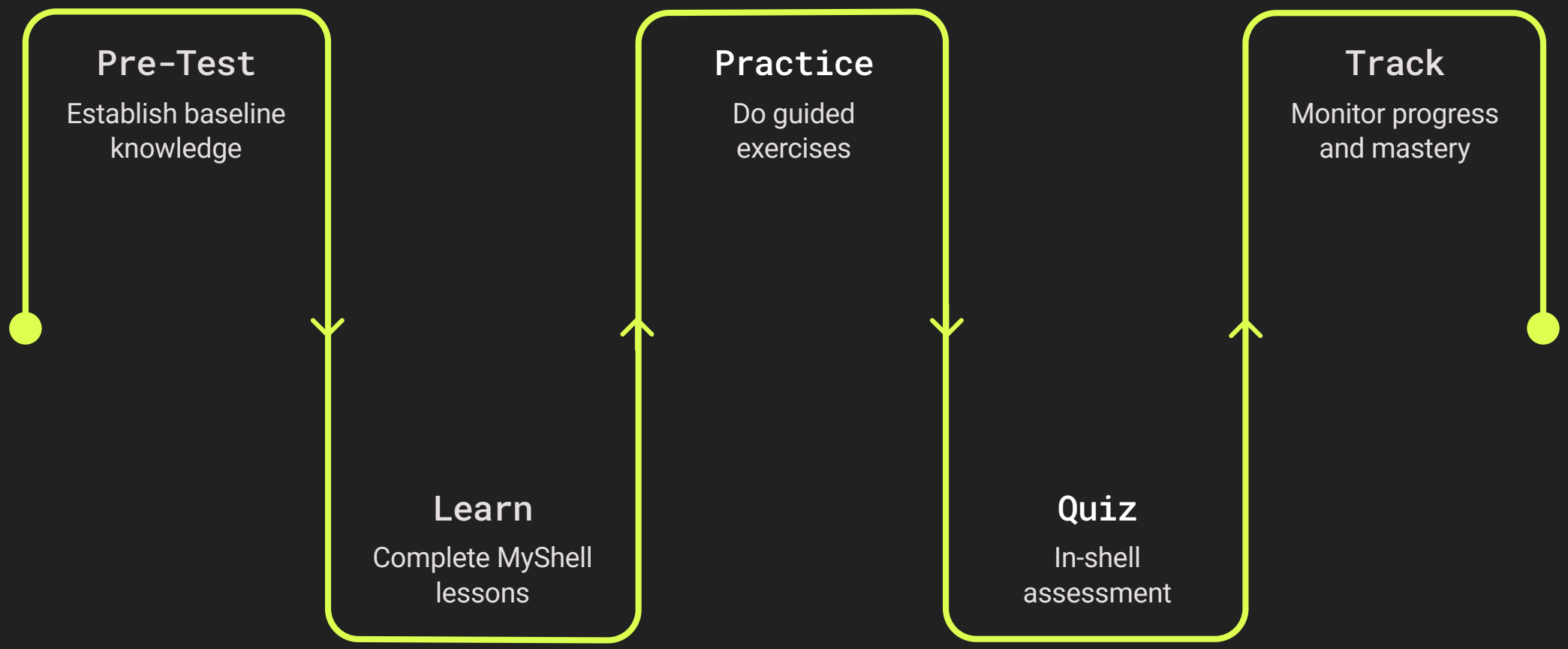
- Beginner-friendly explanations & step-by-step guidance
- Structured practice activities
- Quiz-based assessment with scoring
- Progress tracking and visualization
- Fixed learning roadmap

🔭 Future Scope: Adaptive Learning

- **Difficulty Adjustment:** Adapt lesson complexity based on quiz performance
- **Personalized Exercises:** Practice activities tailored to student level
- **Weak-Area Detection:** Identify commands where students struggle
- **Personalized Learning Paths:** Recommend lessons based on performance gaps
- **Adaptive Recommendations:** Suggest next steps based on mastery level

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Methodology Evaluation Framework



1

Learning Gain

Pre-test vs. post-test score improvement

2

Quiz Performance

Average quiz scores and mastery rate

3

Command Mastery

Commands successfully learned

4

Completion Rate

Practice activities completed

5

Efficiency & Satisfaction

Time per lesson and student feedback

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Accessibility & Dissemination



Distribution & Access

- **GitHub Repository:** Open-source for students and educators
- **Platform:** Linux/Ubuntu — compile with GCC, run from terminal
- **Lightweight:** Minimal resources, suitable for any machine

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Beginner-Friendly by Design

- Clear, simple language throughout — explicit "what to type" guidance
- Text-based interface, no complex GUI required
- Built-in help system and step-by-step progression

Learning Contexts

- Classroom use with instructor guidance
- Self-paced independent study
- Supplementary tool for OS/Linux courses
- Extensible with additional lessons and commands

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Live Demonstration: MyShell in Action

```
MyShell> 1

=== LESSON: ls ===

What is ls?
  ls lists files and directories
  in the current directory.

When to use it:
  Use ls when you want to see
  what files and folders exist.

Example:
  $ ls

What happens inside Linux:
  fork() → exec(ls) → wait()
  OS provides directory listing.

--- PRACTICE ---
Try typing: ls

--- QUIZ ---
What does ls do?
  A) Lists files and directories
  B) Creates a new file
  C) Removes a file
Your answer: A

Score: 1/1 ✓

--- PROGRESS ---
ls      [Mastered ✓]
pwd     [In Progress...]
cd      [Not Started ]
```

What This Demonstrates

1 Integrated Lesson

Command explanation, OS concept, and example — all in one unified view.

2 Hands-On Practice

Students execute the actual command immediately within the learning environment.

3 Instant Assessment

Quiz and automated scoring deliver real-time feedback with no external tools.

4 Visible Progress

Persistent mastery tracking shows exactly where the student stands on their roadmap.

The shell is the classroom. Learn, practice, quiz, and track — all in one terminal session.

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